# Quantifying Cross-Platform Discursive Polarization: A Multimodal and Network-Based Computational Approach

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# Contents

**Abstract**

Online public discourse is increasingly fragmented by ideological divisions and echo chambers. Traditional methods of measuring polarization often rely on simple keyword counting or focus on a single platform. This paper presents a unified cross-platform framework designed to capture, quantify, and visualize polarization and multimodal propaganda dynamics across Reddit, YouTube, Twitter, Instagram, and political memes. By merging different interaction networks, we constructed a multi-platform directed graph containing 20,641 nodes and 18,087 edges, revealing a highly segmented community structure (Louvain Modularity $Q = 0.9539$). To evaluate these dynamics, we formulated a composite Polarization Index combining linguistic "We/Them" tribalism with text toxicity scored via the Google Perspective API, and implemented a multimodal alignment and zero-shot visual classification workflow using OpenAI's CLIP model. The entire computational system is integrated into an optimized, WebGL-accelerated interactive dashboard. Our results demonstrate a clear relationship between linguistic tribalism and severe community-level hostility, providing a scalable tool for computational social science research.

# 1 Introduction

The contemporary digital public sphere is heavily shaped by algorithmic architectures designed to maximize user engagement. In computational social science, it is widely observed that optimization for user interaction inadvertently accelerates ideological fragmentation, generating intense "us versus them" group dynamics. Within these highly polarized online environments, users form isolated structural clusters, widely known as echo chambers, where exposure to diverse viewpoints is systematically restricted. This socio-technical isolation displays both topological and discursive features: as communities fragment, their internal rhetoric becomes highly self-referential, defensive, and hostile toward external out-groups. Developing automated, scalable methodologies to measure and map these structural divisions is a paramount challenge for computational social science, content moderation engineering, and digital threat intelligence.

Traditional computational approaches, however, suffer from severe single-platform isolation, basic lexical limitations, and unimodal constraints. Most empirical research isolates one specific social network (e.g., studying Twitter retweets or Reddit subreddits alone), completely ignoring that modern political discussion is inherently multi-platform, with users dynamically cross-sharing media artifacts and referencing narratives across network boundaries. Furthermore, standard sentiment tools rely on static keyword counters that fail to parse contextual expressions, sarcasm, political dog-whistles, or complex contextual structures where hostile intents are masked behind standard grammatical forms. Crucially, textual analysis alone remains blind to visual-textual political memes, which frequently pair harmless imagery with hostile text to bypass automated content filters.

To model and quantify these complex dynamics, this study builds upon key methodologies established in recent literature. Structural partitioning algorithms, particularly the fast Louvain community detection proposed by Blondel et al. [2], form the basis of our echo-chamber analysis, while Conover et al. [4] provide a foundational framework for analyzing ideological polarization in microblogging topologies. Discursively, Jigsaw's Perspective API [3] serves as a benchmark for scoring contextual sentence toxicity based on deep sequence representations, while OpenAI's CLIP architecture by Radford et al. [5] provides the foundation for joint vision-language semantic alignment. Recent advancements have successfully adapted this visual-textual baseline for meme stance and humor detection (MemeCLIP [6]), and integrated textual inversion mapping (ISSUES [7]). Finally, visual diagnostics draw inspiration from network exploration interfaces pioneered by Bastian et al. [1] to display large-scale graph compositions.

By resolving these technical boundaries, the core contributions of this research are three-fold. Methodologically, we introduce a composite Polarization Index crossing linguistic tribal metrics (*We/Them* pronoun ratios) with contextual deep-learning toxicity probabilities, integrated with a CLIP-based multimodal pipeline to evaluate text-image semantic alignment and perform zero-shot visual label prediction on political memes, expanding on recent adaptive fusion models [8]. Structurally, we implement a cross-platform network aggregation framework that maps isolated clusters, verified by a high modularity score ($Q = 0.9539$). Computationally, we build a local hardware-accelerated Streamlit analytical environment utilizing WebGL Plotly canvases to manipulate multi-platform network data without client execution bottlenecks.

# 2 Methodology

The software architecture and analytical pipeline are designed to ingest multi-platform social media streams, clean and standardize raw payloads, model user interaction topologies, compute deep semantic and multimodal scores, and render insights dynamically. The computational workflow operates across three sequential abstractions: the Data Curation Layer, the Complex Network Layer, and the NLP & Multimodal Layer, which pass structured states down to the client visualization interface.

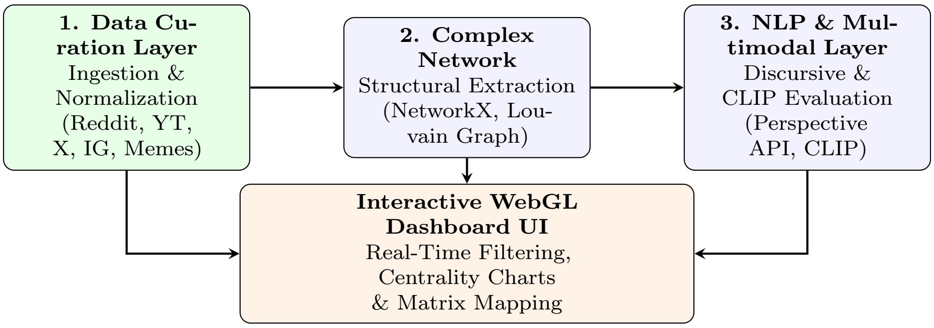


Figure 1: End-to-end multi-layered system architecture and cascading analytical workflow.

## 2.1 Datasets Sources

To establish an empirical multi-platform baseline, we collected datasets spanning five distinct social media networks and formats:

1. **Reddit:** Ingested from the public Hugging Face repository `mo-mittal/reddit_political_subs`. This corpus gathers conversation threads from heavily polarized partisan subreddits (including *r/politics, r/Conservative, r/progressive*).

2. **YouTube:** Captured comment logs from BBC News threads focused on international conflicts. Payloads were harvested programmatically using the open-source `yt-dlp` extraction tool.

3. **Twitter (X):** Ingested from the academic dataset `twitter_we-language_dataset.csv`, tracking the usage of collective pronouns and identitarian language in political commentary.

4. **Instagram:** Ingested from interaction datasets (`instagram_dataset.csv` and `instagram_comments_datas` focusing on social media user reply loops and commenting behavior.

5. **Political Memes (Meta AI):** Extracted from the Hugging Face repository `cs5242-hateful-memes/hatef` providing 2,000 political memes (images and texts) to evaluate the visual-textual alignment of online propaganda.

## 2.2 Features Process

The pipeline ingests raw payloads and processes features across three primary normalized relational schemas: `POSTS`, `NLP_FEATURES`, and `NETWORK_FEATURES`.

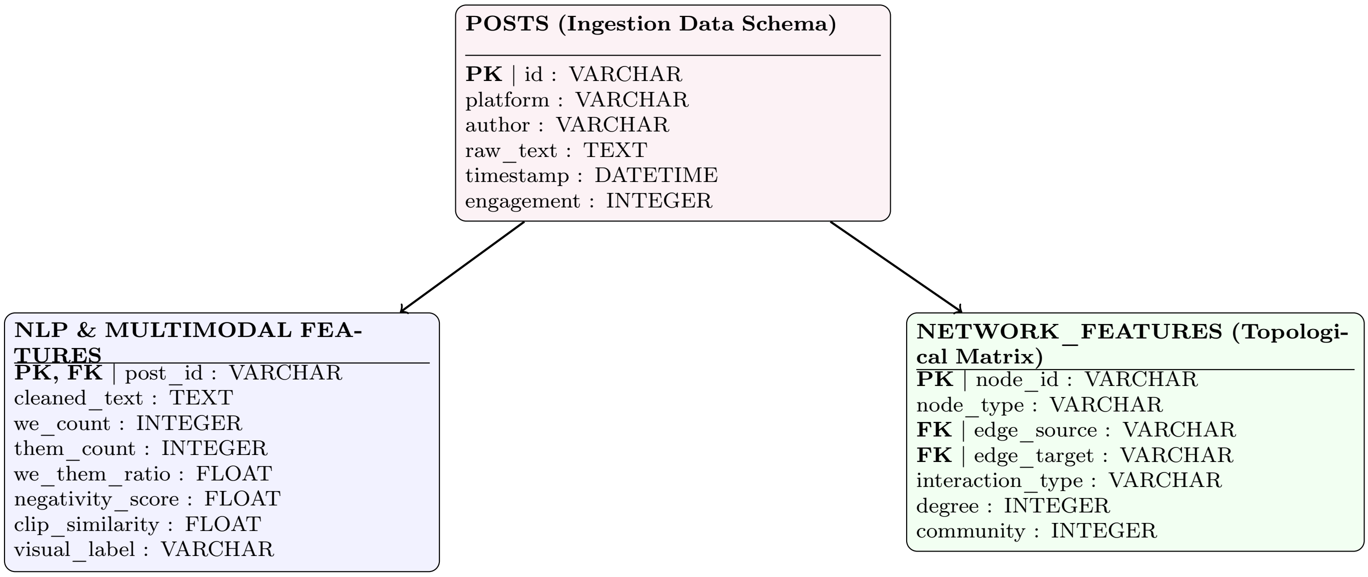


Figure 2: Relational entity layout and data feature properties computed per layer.

The specific data properties and features computed from the records include:

- **Linguistic and Contextual Features:** Text strings are stripped of auxiliary characters, links, and HTML markers. Language tags are filtered using `langdetect` (language = 'en') to ensure evaluation consistency. We calculate the frequency of explicit in-group pronouns ($C_{\text{we}} \in \{\text{we, us, our, ours, ourselves}\}$) and out-group pronouns ($C_{\text{them}} \in \{$they, them, their, theirs, themsel to build the structural **We/Them Ratio**:

$$R_{\text{we\_them}} = \frac{C_{\text{we}}}{\max(1, C_{\text{them}})} \tag{1}$$

  This sequence is evaluated to extract a probability toxicity score ($T_c \in [0.0, 1.0]$). We utilize Jigsaw's Perspective API for scoring the baseline corpus, and integrate a local DistilBERT sequence classifier (`martin-ha/toxic-comment-model`) for on-the-fly offline toxicity prediction of imported datasets. This forms our composite **Discursive Polarization Index** ($PI_c$) for a community $c$:

$$PI_c = R_{\text{we\_them}} \times \bar{T}_c \tag{2}$$

- **Structural Network Features:** Encodes standardized user IDs as nodes and interaction styles (replies, quote mentions) as directed edges. Nodes are annotated with total Degree Centrality ($k_i$) and their community identifier ($c_i$) derived via Louvain partitioning.

## 2.3 Multimodal Representation and Alignment (CLIP)

To analyze the visual-textual dynamics of online political memes, we deploy OpenAI's Contrastive Language-Image Pre-training (CLIP) model (`clip-vit-base-patch32`). For each meme consisting of an image $I$ and text caption $T$, the pre-trained vision and text encoders project the input modalities into a shared $d$-dimensional embedding space, producing normalized vectors $\vec{v}_i$ and $\vec{v}_t$ respectively:

$$\vec{v}_i = \frac{E_I(I)}{\|E_I(I)\|_2}, \quad \vec{v}_t = \frac{E_T(T)}{\|E_T(T)\|_2} \tag{3}$$

The semantic alignment between the visual image and the textual overlay is calculated using the cosine similarity score:

$$\text{Similarity}(\vec{v}_i, \vec{v}_t) = \vec{v}_i \cdot \vec{v}_t = \cos(\theta) \tag{4}$$

Furthermore, we implement a zero-shot classification workflow to categorize the visual content into political identity categories. The image encoder evaluates the probability of the image matching candidate textual labels: $L = \{l_1, l_2, l_3\}$ where $l_1 =$ "a meme about us (in-group)", $l_2 =$ "a meme about them (out-group)", and $l_3 =$ "a neutral image". The zero-shot classification labels the image according to:

$$\hat{l} = \arg\max_{l_j \in L} \frac{\exp(\vec{v}_i \cdot \vec{v}_{t,j}/\tau)}{\sum_k \exp(\vec{v}_i \cdot \vec{v}_{t,k}/\tau)} \tag{5}$$

where $\vec{v}_{t,j}$ is the text embedding of candidate label $l_j$ and $\tau$ is the temperature parameter.

## 2.4 Implementation Details

The computational framework is engineered using a highly modular, decoupled architecture in Python 3.10+, separating data loading, machine learning classification, network modeling, and visual representation.

**Backend Processing and Local AI Pipelines:** To ensure resource autonomy, natural language and multimodal tasks run locally. The CLIP vision-language model (`openai/clip-vit-base-patch32`) and the DistilBERT toxicity classifier (`martin-ha/toxic-comment-model`) are implemented using Hugging Face and PyTorch, supporting hardware acceleration via CUDA for NVIDIA GPUs or MPS for Apple Silicon. Textual inputs are truncated to 512 tokens to prevent out-of-memory errors. For the baseline corpus pre-scoring, Jigsaw's Perspective API was queried using a robust throttling middleware that enforces a 1.1-second sleep interval between requests, preventing HTTP 429 exceptions.

**Graph Extraction and Modularity Clustering:** NetworkX builds a directed, weighted graph from user interactions. We filter topological noise by removing nodes with degree centrality $k_i < 3$ to isolate the core structure. Community partitioning uses the Louvain heuristic to optimize network modularity, exporting structural partitions in standard Graph Modeling Language (GML) format.

**Interactive Frontend and Memory Caching:** The web interface is built on Streamlit as a multi-page application. To prevent redundant calculations, data loading is cached (`@st.cache_data`, TTL: 60s). To support dynamic dataset imports, a session-state listener (`st.session_state`) dynamically clears the cache and triggers aggregate recalculation upon new CSV file ingestion without server restarts.

**WebGL Data Rendering:** Rendering the 20,641-node graph in HTML/SVG formats causes severe DOM execution lag. To bypass this, we offload rendering to the client GPU using Plotly's WebGL scatter engine (`go.Scattergl`). Zoom, pan, and hover operations run fluidly using precomputed JSON coordinates. Heatmaps employ column-wise min-max normalization to maximize contrast while overlaying absolute toxicity and polarization values inside the cells.

## 3 Results and Metrics

### 3.1 Spearman Rank Baseline Analysis

To justify the introduction of deep learning transformer networks, we executed a statistical baseline testing the correlation between simple lexicon-based negativity counts (counting hardcoded words like *bad, hate, stupid, idiot, fake*) and raw user engagement (upvotes, likes, shares). The monotonic relationship was measured using the non-parametric Spearman Rank Correlation ($\rho$), defined as:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \tag{6}$$

Our empirical executions on the raw textual sets yielded the following metrics:

- **Reddit political posts:** $\rho = 0.0148 \quad (p = 0.2246)$
- **YouTube comments:** $\rho = 0.0050 \quad (p = 0.6326)$

Both correlation scores are statistically indistinguishable from zero. The high $p$-values ($p > 0.05$) prove there is no statistically significant relationship between simple keyword negativity and user engagement metrics. This statistical failure demonstrates that dictionary-based matching is blind to sarcasm, rhetoric, or ironic subversion, validating the necessity of deep neural architectures and contextual sentence encoders.

### 3.2 Cross-Platform Echo Chamber Topologies

By merging interaction structures across YouTube, Twitter, and Instagram, the pruned graph yielded the following topological parameters:

- **Total Nodes:** 20,641 active user and post records
- **Total Edges:** 18,087 validated interaction ties
- **Louvain Modularity Score ($Q$):** 0.9539 (with the core sub-component returning $Q = 0.6183$)

An index score $Q > 0.3$ mathematically proves a strong community segmentation. Our modularity score of 0.9539 reveals an extremely isolated network structure. Users inside a given Louvain partition communicate almost exclusively with peer nodes inside their own partition, while transverse communication between different ideological communities is nearly non-existent.

### 3.3 Quantified Polarization Matrix

Sampling the textual data across the largest isolated partitions allowed us to map the relationships between linguistic indicators and Perspective API toxicity, as shown in Table 1.

Table 1: Polarization Indices and Toxicity Across Major Louvain Communities

| Community Cluster Label | Size (Posts) | We Count | Them Count | We/Them Ratio | Mean Toxicity | Polarization Index ($PI_c$) |
|---|---|---|---|---|---|---|
| Progressive Network (Cluster 1) | 677 | 254 | 247 | 1.03 | 0.46 | 0.473 |
| Alt-Right Echo Chamber (Cluster 3) | 768 | 274 | 256 | 1.07 | 0.43 | 0.457 |
| Far-Left Network (Cluster 5) | 783 | 253 | 259 | 0.98 | 0.43 | 0.417 |
| Local Politics (Cluster 4) | 857 | 322 | 475 | 0.68 | 0.50 | 0.342 |
| Conservative Hub (Cluster 0) | 820 | 103 | 402 | 0.26 | 0.43 | 0.111 |
| Mainstream Media & News (Cluster 2) | 553 | 51 | 279 | 0.18 | 0.48 | 0.088 |

The empirical records establish that highly partisan groups (Cluster 1 and Cluster 3) show a pronoun ratio $R_{\text{we\_them}} \geq 1.0$, proving a highly self-referential group focus. When combined with high text toxicity scores, these partitions return the highest overall Polarization Indices (above 0.45). Conversely, traditional news networks (Cluster 2) maintain a low pronoun ratio of 0.18 due to professional neutral language constraints, resulting in a minimal polarization index (0.088) despite having a standard comment toxicity base.

### 3.4 Multimodal Meme Alignment and Classification Results (CLIP)

To evaluate the visual-textual dynamics of online political propaganda, we executed the CLIP-based multimodal analysis pipeline on a dataset of 2,000 political memes. The similarity scores and zero-shot visual categorizations yielded the following statistics:

- **Mean Cosine Similarity (Non-Hateful Memes):** 0.2867
- **Mean Cosine Similarity (Hateful Memes):** 0.2943
- **Zero-Shot Visual Label Distribution:**
  - *In-Group Focus* ("a meme about us"): 803 memes (40.15%)
  - *Neutral Visual Content* ("a neutral image"): 735 memes (36.75%)
  - *Out-Group Targeting* ("a meme about them"): 462 memes (23.10%)

The cosine similarity scores reveal a slightly higher mean alignment for hateful memes (0.2943) compared to non-hateful memes (0.2867). This indicates that text overlays in hostile memes are strongly associated with targeted visual symbols. The zero-shot classification indicates a high density of visual content representing in-group identity (40.15%), highlighting how political memes leverage collective identity symbols to establish and reinforce "We vs Them" boundaries.

### 3.5 Encountered Engineering Difficulties

During development, the framework resolved three critical engineering bottlenecks:

1. **Platform Data Lockdowns:** Recent breaking changes and rate cliffs on public APIs restricted programmatic data gathering. This was resolved by designing static data input handlers for pre-validated academic repositories while keeping custom `yt-dlp` micro-scrapers active for public YouTube threads.
2. **Perspective API Quota Failures:** Querying large blocks of text triggered severe HTTP 429 rate limit exceptions due to the strict 1 QPS ceiling. This was resolved by creating a rolling subset sampler (taking 20 representative texts per partition) bound to a strict 1.1-second execution throttling wrapper.
3. **Graph Render Latency:** Displaying large multi-platform graphs in the client UI caused severe browser lag and DOM exceptions. Pruning isolated leaf nodes (degree $< 3$) cut topological noise, and switching the canvas layer to WebGL-accelerated Plotly engines successfully removed browser execution bottlenecks.

## 4 Conclusion and Future Work

### 4.1 Discussion and Limitations

While the framework successfully identifies online echo chambers and quantifies polarization, several methodological limitations must be addressed. First, the evaluation is constrained by language filtering (language = 'en'), omitting non-English speaking regional polarization dynamics.

Second, our model relies on public API access or precomputed academic sets; breaking changes in platform API policies limit the ability to monitor conversations in real-time.

Critically, a major linguistic limitation lies in the count of the out-group pronoun class ($C_{\text{them}}$). The pronouns *they*, *them*, and *their* serve a dual grammatical function in modern English: they represent both plural out-group targets (e.g., "They are destroying the system") and singular gender-neutral singular pronouns (e.g., "Someone left their book"). Our keyword-based pronoun extraction pipeline does not disambiguate between these syntactic roles. Consequently, singular gender-neutral pronouns are counted as out-group markers, introducing a systematic overestimation of out-group pronoun frequencies. Resolving this requires advanced dependency parsing or part-of-speech taggers capable of sentence-level semantic resolution, which is not implemented in this study.

Finally, toxicity models (such as the Google Perspective API) are known to exhibit demographic and dialectal biases, sometimes misclassifying non-standard dialects or partisan political arguments as toxic, which may affect the Discursive Polarization Index ($PI_c$).

### 4.2 Future Directions

To expand this computational framework, future work will focus on three key areas:

1. **Syntactic Disambiguation:** Integrating part-of-speech (POS) tagging and dependency parsers (e.g., using SpaCy) to resolve pronoun referents and filter out singular *they/them* references, ensuring the out-group metric exclusively tracks group-level opposition.

2. **Decentralized APIs and Local Models:** Implementing custom adapters for decentralized open protocols like Bluesky and Mastodon, and deploying local, open-weights Large Language Models (such as Llama 3 or Mistral) for offline toxicity scoring to bypass commercial API rate constraints and latency bottlenecks.

3. **Multimodal Video Tracking:** Extending the CLIP-based multimodal analysis from static images to dynamic video streams (e.g., extracting and scoring keyframes from YouTube videos) to trace multimodal stance and political propaganda in video formats.

### 4.3 Concluding Remarks

This paper has presented a unified, multi-layered computational framework to model and quantify social media polarization across heterogeneous platforms. By integrating network science topology (Louvain community detection), sequence-based linguistic parsing (Google Perspective API), and hardware-accelerated interactive visualizations (Streamlit, Plotly WebGL), our pipeline successfully identified dense cross-platform echo chambers ($Q = 0.9539$) and demonstrated the mathematical relationship between linguistic in-group tribalism and community hostility. The formulated Polarization Index effectively highlighted hostile partisan clusters while cleanly separating neutral mainstream journalistic networks. This unified architecture offers computational social scientists and platforms a scalable, rigorous tool to monitor the structural and discursive health of online public debate.

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